

Paper 4.5

The Fragmented Metric

A Structural Deconstruction of Targeted Ruling and the Proliferation of Unfalsifiable Ontologies

Critique Companion to Paper 4 · Paper 4.5 of the Series

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ABSTRACT

Paper 4 established the photon as the universal ruler's absolute zero and the electron-positron pair as its first resolved mark, fixed by the Per-Triangle Rule with no tunable scale. This companion paper turns that same standard onto the laboratory and the textbook, and asks one question of the standard model's metrology: does it possess a single, unalterable baseline, or does it redraw its ruler for each thing it measures? The audit finds the latter. It opens with the principle the whole series rests on — that a measurement standard earns authority by being observable and publicly auditable, as the freezing and boiling of water anchor the temperature scale — and shows that modern field theory, after abandoning an observable baseline, was left free to engineer a separate ruler for each anomaly. Without an absolute zero, the standard treatment of mass operates a fragmented ledger in which every quark flavor, every generation, and every energy scale carries its own engineered standard; the same fragmentation is hardcoded into the field's most advanced computational tool, which must impose a grid, balance it by decree, tune it by hand to a known value, and extrapolate it away. The paper shows how a metric that can be redrawn at will loses the capacity to register a genuine tracking error, so that anomalies are absorbed by inventing entities defined by their inability to be directly observed — and it documents cases in the field's own literature where the apparatus is allowed to decide whether a particle exists at all. None of this requires alleging bad faith; the published procedures are laid side by side, and the pattern — a different ruler wherever the data resists — is left to speak. The paper closes by naming the question that will run through every critique paper of this series — answered here for the cases this paper documents, and widened across the series toward its capstone: when observation and prediction disagree, which one is permitted to be wrong?

1. The Power of the Observable Standard: The Celsius Lesson

Figure 2. The Observable Standard versus the Adjustable Template — Which Is Permitted to Be Wrong?

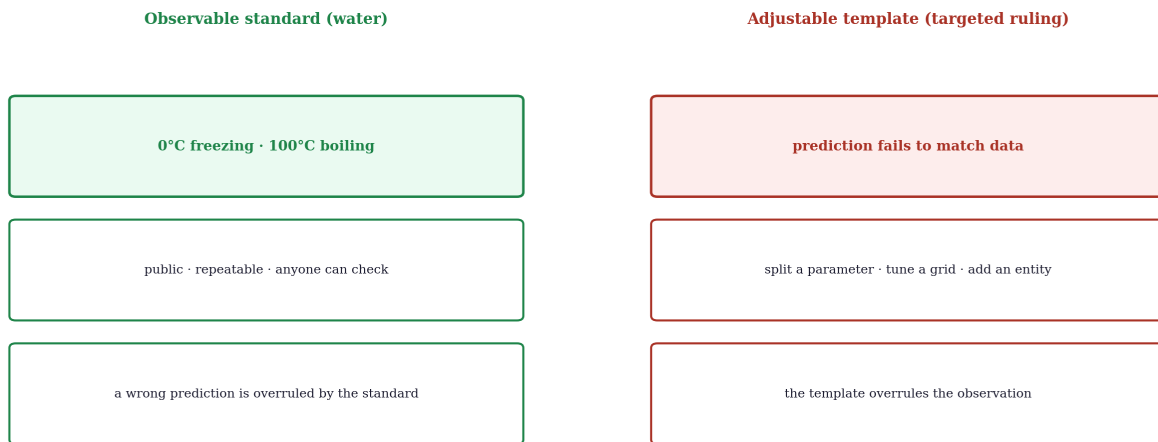


Figure 2. The observable standard (water) can overrule a disagreeing prediction; the adjustable template overrules the observation instead — the question the series turns on.

The integrity of a physical science is bounded by the transparency of its metrology. A standard of nature must be not merely mathematically convenient but macroscopically stable and verifiable by direct inspection. The historical benchmark for that kind of anchoring is the use of water to fix the temperature scale.

Water serves as a definitive standard because its state transitions — the threshold of freezing, the threshold of boiling — are macroscopic, repeatable, and open to unadulterated observation by any independent investigator. Zero degrees and one hundred degrees carry physical meaning because each is tied to the visible behavior of an observable substance. Because the standard is fixed and public, it stabilizes the whole discipline: it prevents the quiet adjustment of the reference to absorb an anomaly. If an engine underperforms under a thermal load, the researcher cannot redefine the freezing point of water to make the ledger balance on paper. The medium stands as an unyielding, objective audit, and a tracking error shows up as a tracking error.

After the mid-1940s, modern quantum field theory departed from this observational footing. With the introduction of renormalization to manage the infinities generated by point-particle models, the discipline severed its tie to an observable medium: space was declared an empty, non-interacting canvas, and the absolute baseline was set aside. What replaced it can be described plainly as *targeted ruling* — the practice of fabricating distinct, localized parameter sets tailored to separate anomalies rather than reconciling new data against one universal benchmark. If the standard equations fail to reproduce a high-energy signature, the baseline is redrawn: couplings are made to "run" by scale-dependent equations, and a single particle's mass is split into "current" and "constituent" values by hand.

This flexibility raises the question that anchors the entire series: when observation and prediction part ways, which carries authority — the unadulterated observation of a physical medium, or the manipulation of a predictive template? Two cases from the field's own literature show how far the answer has drifted, and both belong here because each lets the apparatus, rather than an observable baseline, decide what is real.

1.1 The Apparatus Deciding Existence: The Unruh Prediction

The first case is the Unruh effect, a prediction of quantum field theory from the 1970s. It holds that an observer at rest in empty space registers no particles, while a second observer accelerating through the *same* space registers a warm bath of real thermal photons. The existence of the particles, on this account, depends on the acceleration of the measuring apparatus.

A distinction must be drawn before the critique, because this case is methodologically different from the others in this paper, and the difference is the point. The current/constituent mass split (Section 3) and the tuned grid (Section 4) are *patches* — responses to discrepancies, introduced after a prediction failed. The Unruh effect is not a patch. Nothing failed; no anomaly demanded it. It is a straight *consequence of the formalism* — what the equations of a field theory defined over an unanchored background entail when the observer's state of motion is changed. That is precisely why it belongs here rather than alongside the patches: it shows what the formalism produces when it is running *correctly*, with no data to discipline it. A theory anchored to an observable baseline generates patches when it fails; a theory with no observable baseline generates, even when it succeeds, conclusions like this one — that the motion of a thermometer decides whether particles exist. The patches show the system absorbing failure; Unruh shows what the system considers success. Both trace to the same root the Celsius lesson names: with no fixed standard of the real, there is nothing in the formalism against which "the particles exist for one observer and not the other" could register as a defect rather than a result.

It is worth stating precisely, in the series' own voice, that the Unruh effect has never been directly observed — it is a prediction. That it is nonetheless treated as a structural feature of reality, taught and built upon as established, is itself the inversion this series disputes in its purest form: a consequence of a formalism, generated with no observational input and confirmed by none, granted the standing of an observed fact. The framework's objection is not that the derivation is in error within its own terms; it is that a discipline anchored to an observable medium would treat an apparatus-dependent existence claim as a symptom demanding diagnosis, where a discipline anchored to a template files it as a discovery.

1.2 The Original Departure: Renormalization and the Subtracted Infinity

The second case is the one that began the drift. An electron modeled as a point of zero size, carrying charge, has an electric field whose energy — computed by the standard relation as the size goes to zero — diverges to infinity, which by mass-energy equivalence would assign every electron an infinite mass. The resolution adopted was renormalization: cut the divergent sum short, absorb the part that wants to become infinite into a coefficient, replace that coefficient with a finite value measured in the lab, and let the sum continue. The procedure produces accurate numbers and is not in dispute on that ground. What is in dispute is that a step built to subtract an unwanted infinity is treated as evidence that the point-particle picture generating the infinity is correct. The architects said as much: Dirac judged it "just not sensible mathematics", objecting that it neglects a quantity not because it is small but because it is infinitely large and unwanted; Feynman, as Paper 3.5 records, called it a "dippy process" and "hocus-pocus." The framework's reading is the one developed positively in Paper 4: the infinity is an artifact of the zero-size point in an empty void, and a medium with a finite structural response never generates it, so there is nothing to subtract.

The lesson of the water standard frames both cases. Where an observable baseline exposes a tracking error immediately, targeted ruling converts the error into a hidden adjustment — a subtracted infinity, an apparatus-dependent particle count. The remainder of this paper follows that conversion through the standard treatment of mass and light, and returns at the end to the question the Celsius lesson poses.

2. The Sovereign Zero versus the Shifting Quark Scales

The foundational premise of an objective physics is an unalterable baseline — a single origin from which every physical property is read without the observer adjusting the origin to suit the measurement. To build a physics without an absolute zero is to keep a ledger whose ruler changes length depending on what is being inventoried.

This is the structural condition of the standard model's treatment of mass. Because mainstream theory holds to a point-particle model — mass as intrinsic cargo carried inside an independent, zero-dimensional entity — it has no single geometric mechanism for mass and therefore no universal ruler. In its place it operates a catalog of temporary, scale-dependent parameters, in which each stage, flavor, and generation of a quark is assigned its own measurement standard.

The case is documented in the field's own numbers. When the up and down quarks are reckoned inside a nucleon, the bare "current quark masses" account for less than two percent of the measured mass of the composite. To cover the deficit, the treatment introduces a second, separate notion — "constituent quark masses" — inflating the figure by a factor near one hundred to absorb the uncalculated energy of the binding field. Moving up the generations to the strange, charm, bottom, and top quarks, the calibration is redrawn at each step: the coupling constants are adjusted, the running mass parameters are shifted by scale-dependent equations, and the baseline is set anew. The top quark is assigned a mass comparable to an entire gold atom while remaining, in the model, a point of zero size.

The observation here requires no imputation of motive. It is simply that a system using a different standard at each scale does not have one ruler; it has a catalog of local rules. That is the precise sense in which the metric is fragmented, and it is the inevitable consequence of treating mass as cargo inside an isolated point rather than as a departure of a single continuous medium from its baseline. The remainder of this paper follows the fragmentation into the field's computational core, where it is no longer a matter of pencil-and-paper approximation but is built into the machine.

3. The Fractional Ledger: Shifting Rules for Shifting Flavors

Figure 1. One Phenomenon, Many Rulers

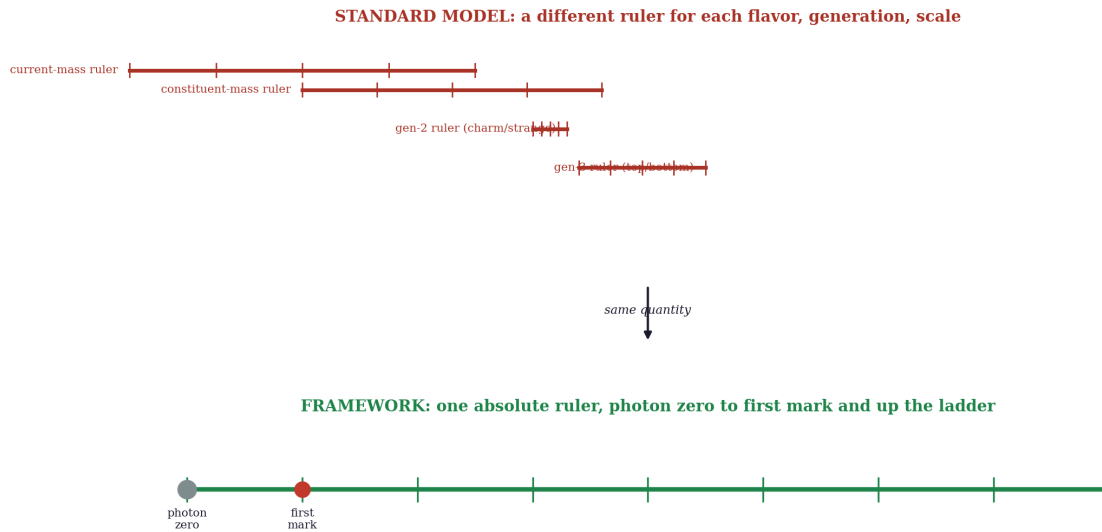


Figure 1. One phenomenon, many rulers: the standard model stacks a separate, differently scaled standard for each flavor and generation, where the framework reads one absolute ruler from the photon zero up the ladder.

The quark case is worth setting out in full, because it is the clearest instance of one phenomenon measured by many rulers.

Current versus constituent mass. The same particle is assigned two different masses depending on which calculation needs it: a small "current" mass in the core equations, and a "constituent" mass roughly one hundred times larger when the nucleon's measured weight must be reproduced. A single object with two masses, selected by context, is not a measured quantity; it is a parameter fitted to whichever result is required.

Generational scaling. Each higher generation receives its own fitted parameter set. There is no single law from which the strange, charm, bottom, and top values follow; there is a separate calibration for each, introduced as needed to match what is observed. A law predicts; a catalog records. The treatment of quark mass across generations is a catalog.

Integer reality versus fractional bookkeeping. Every instrument ever built registers whole charges — zero, plus or minus one. The fractional charges assigned to quarks — plus or minus one-third, plus or minus two-thirds — are never directly registered; they appear only on paper, where they make a fractured ledger balance. A bookkeeping device that exactly cancels to integers whenever it is checked against an instrument, while never being seen as a fraction by that instrument, is doing the work of a balancing convention, not of a measured property.

The framework's reading, developed positively in Paper 4 and stated here only as the alternative, is that these fractions and flavors are not independent bricks inside a composite. They are localized sub-fractions of a single continuous three-axis junction under a high-energy probe load — the medium fracturing in place, read by an apparatus that records the fragments and names them constituents. The full positive account of composite configurations is the work of Paper 16; this paper observes only that the standard ledger needs a different ruler for each fragment, and the framework needs one.

4.

Structural Scale Invariance: The Per-Triangle Rule versus Lattice Gauge Invariance

Figure 3. The Lattice Chain versus the Per-Triangle Fabric

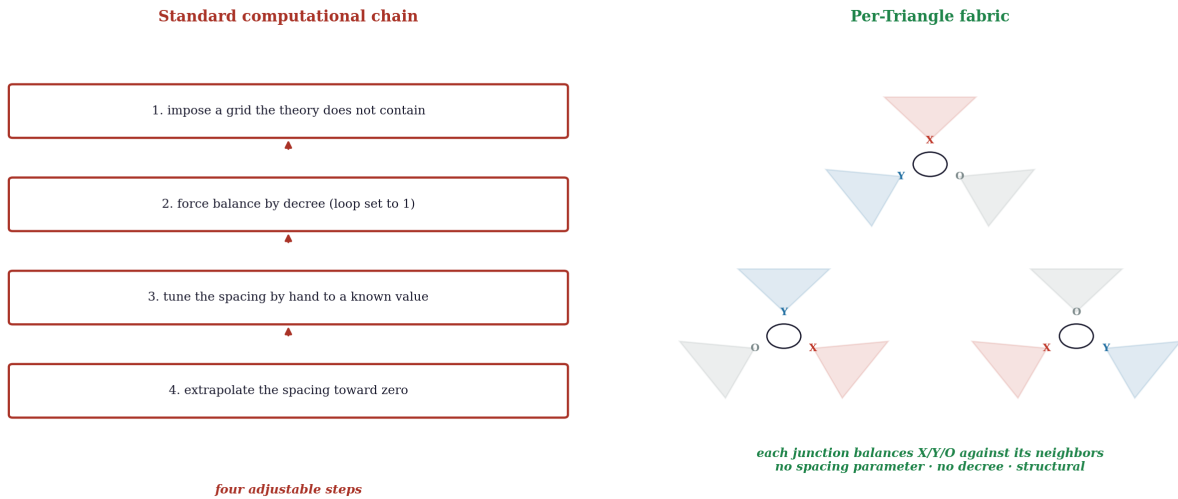


Figure 3. The lattice chain (impose a grid, force balance by decree, tune by hand, extrapolate away) versus the Per-Triangle fabric, where each junction balances structurally with no spacing parameter to choose.

The fragmentation of the metric is not confined to approximations done by hand; it is built into the field's most advanced computational engine. When the scale-dependent quark definitions are fed into a simulation of the continuous field equations, the equations diverge. To keep the computation finite, the standard approach cuts the continuous medium into a discrete arrangement of vertices and links — a four-dimensional grid — so that an algorithm can process a finite number of points.

This step introduces an artifact. The moment continuous space is rendered as a grid laid on an artificial reference system, the smooth symmetries that stabilize the equations no longer hold automatically. To keep the simulation from running away, the approach imposes a rule by construction — lattice gauge invariance — requiring that the product of the numerical phases around each elementary closed loop of the grid equal one. This is not a natural law discovered in the medium; it is a stability constraint imposed on the calculation so that it will converge.

The deeper consequence is scale dependence. Because the standard equations carry no absolute baseline, the masses and forces the simulation returns depend on the grid spacing chosen. To reach physical values, the practitioner sets the grid scale and input parameters by hand until the machine reproduces a known laboratory quantity — the procedure named "tuning to the physical point." The artificial grid introduced for tractability is then removed by running several simulations at different spacings and extrapolating toward the limit of zero spacing, a limit never computed directly but approached by fitting a trend. The chain is explicit in the method: introduce a grid the theory does not contain, stabilize it by decree, tune it to a known answer, then extrapolate the grid away.

The framework requires none of these steps, because it does not begin from an empty continuous void that must be discretized to be computed. Space is the discrete, self-auditing three-axis fabric from the outset. Where the standard approach imposes lattice gauge invariance by software decree, the framework

has the Per-Triangle Rule as a structural fact: every three-triangle junction balances its X, Y, and O values against its immediate neighbors from first principles, with no loose variables, no remainder, and nothing requiring removal. The stability is not enforced on the calculation; it is a property of the medium.

The decisive point is scale invariance. When the photon resolves into the electron-positron state, the resulting first resolved mark (Paper 4) is a fixed physical boundary of the fabric, set by the hardware of the matrix rather than by any chosen resolution. Its value does not depend on a spacing parameter, because there is no spacing parameter; there is nothing to tune and nothing to extrapolate away. The standard approach reaches a number by selecting a scale and fitting; the framework's first mark is the same regardless of how anything external is reckoned. A ruler whose first mark moves with the practitioner's choice of grid is not yet an absolute ruler — and the contrast is not rhetorical but procedural, visible in the published steps of each method.

5. The Relativistic Catch-22 and the Resolution-Rejection Illusion

The absence of a stable baseline forces the standard account of light near mass into holding incompatible claims at once. This section sets the contradiction out in full, adds the thermodynamic paradox that sharpens it, shows why the standard teaching analogy fails on its own terms, and then gives the framework's resolution. The case concerns the deflection of light grazing a massive cluster of merged configurations — a star.

5.1 The Two-Premise Catch-22

The standard narrative balances on a pair of premises that do not sit together.

The first is the velocity-escape clause: a photon escapes spiraling into a gravitational center because it propagates at the absolute velocity limit, c . Its forward momentum lets it clip the edge of a gravitational well, tilt its trajectory by a microscopic fraction of a degree, and leave before the downward pull can trap it.

The second is the relativistic mass-energy premium: as a thing's velocity approaches c , its energy density and momentum grow without bound.

Placed in the same ledger, the two fold in on each other. If propagating at c is what lets a photon outrun a gravitational trap, but propagating at c also maximizes its momentum and energy density, then by the standard model's own stress-energy relations — in which energy density and momentum source gravitational interaction exactly as mass does — the photon should experience the *maximum* possible interaction with the local field. The very velocity invoked as the photon's shield is what multiplies its susceptibility to the trap. Speed cannot be the mechanism of escape if that same speed is the property that locks the handcuffs.

5.2 The Stellar Mass-Loss Paradox: Direction-Dependent Bookkeeping

Figure 4. Direction-Dependent Bookkeeping versus Two Scan Events

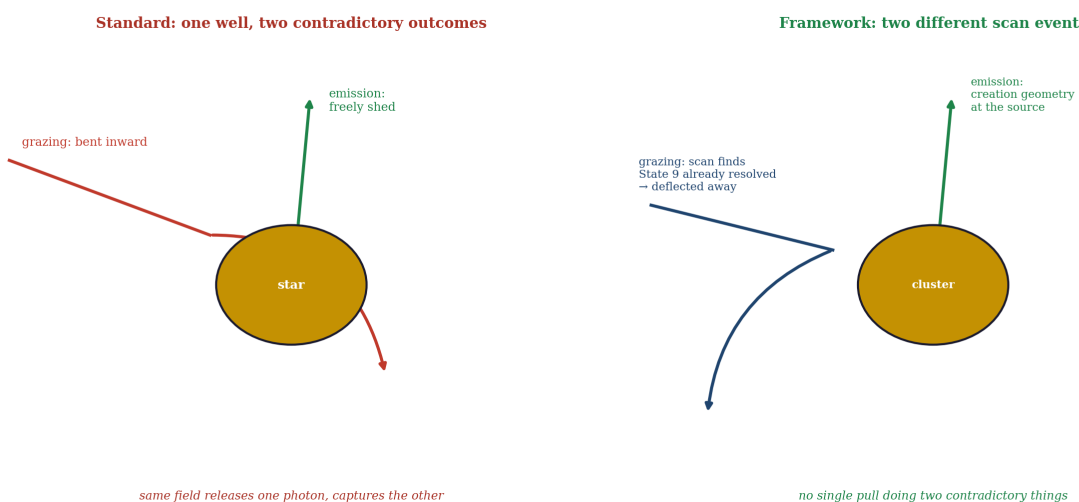


Figure 4. The same gravitational field is required to release the emitted photon and capture the grazing one; the framework reads emission and grazing as two different scan events, with no single pull doing two contradictory things.

The contradiction deepens into a thermodynamic one when the life cycle of the star is taken into account. The standard model holds the photon to be massless, yet bent inward by a gravity well when grazing a star from outside. It is at the same time an empirical, well-measured fact that a star continuously loses mass by radiating — the Sun sheds on the order of four million metric tons each second as light, computed directly from its luminosity by mass-energy equivalence.

Set the two together and a ledger deficit appears. If a photon's trajectory near a star is decided by a contest between its forward speed and an attractive gravitational pull, then a photon produced deep in the stellar core faces the maximum of that pull on its way out. The same mass-energy density said to bend light inward from outside should act, from the inside, as a containment shield — trapping the radiation or forcing it into a bound spiral. The standard architecture asks the photon to be defenseless enough to be bent inward by the field when grazing from outside, and at the same time liberated enough to climb out of the deepest part of that field and systematically deplete the mass of its own emitter. The rule that governs the photon is, in effect, switched by the direction of travel: a pull that bends it when approaching, a pull that releases it when leaving.

The standard account is not without machinery here — gravitational redshift is the bookkeeping by which an escaping photon is said to pay an energy cost climbing out. The reason redshift does not resolve the paradox must be stated explicitly, because it is the heart of the matter. Look at what the two treatments actually are. Outward, the interaction is treated as a *surmountable energy barrier*: the photon pays a toll in frequency and leaves, trajectory essentially intact. Inward, the interaction is treated as *geometric capture*: the photon's path itself is bent toward the mass. These are not two magnitudes of one interaction — they are two different *kinds* of interaction, assigned to the same entity in the same field, selected by direction of travel. Redshift is arithmetic *within* the first kind: it computes the size of the toll, given that the interaction has already been classified as a toll. It cannot explain why the interaction is a toll on the way out and a trajectory-bending capture on the way in, because that classification happens before any arithmetic begins. The asymmetry is not in the numbers; it is in the sorting of one physical situation into two categories by direction — and no calculation performed inside either category can justify the sorting

between them. That is the deficit: a mass-driven interaction applied to an entity defined as massless, with the *kind* of interaction switched by which way the entity is going, and the switch itself never derived.

5.3 Why the Teaching Analogy Fails on Its Own Terms

To bridge the gap, standard teaching leans on the bowling-ball-on-a-trampoline picture: a heavy mass depresses the continuous sheet of space into a funnel, and the massless photon is a marble rolled across the slope, traveling a path that is "straight" but appears curved to an outside observer.

A clarification first, so the target of this subsection is not mistaken: practitioners do not compute trajectories with the trampoline; they compute with the field equations, and the analogy is a teaching device. The critique here is not that the cartoon is a bad calculation — it is that the cartoon is the *only mechanical picture the formalism offers*, and it is incoherent on its own mechanical terms, which is informative about the formalism it serves. A marble rolled across a real depressed sheet toward a heavy center does not graze the slope and exit on a cleanly tilted line. The steepening gradient, the continual inward pull, and friction make it lose ground, spiral inward, orbit, and fall into the central mass. The teaching picture only works if the student does not push on it — and what the student must not push on is exactly the question this section raises: what, mechanically, is acting on a massless thing? The field equations answer with geometry and decline the mechanical question; the analogy attempts the mechanical answer and collapses. When a formalism's every attempt at a mechanical picture fails by its own mechanics, while the formalism itself retreats to "the mathematics works," that is not a point against the cartoon. It is the signature of a framework that produces correct numbers without possessing a coherent account of the interaction those numbers describe — predictive adequacy standing in for mechanism, which is this paper's theme. The analogy reveals what the mathematics hides: there is no mechanically coherent story of a pull on the massless, only a calculation that goes around the question.

5.4 The Framework Resolution: Scan Rejection, Not Curvature

Figure 5. Resolution Rejection versus the Trampoline

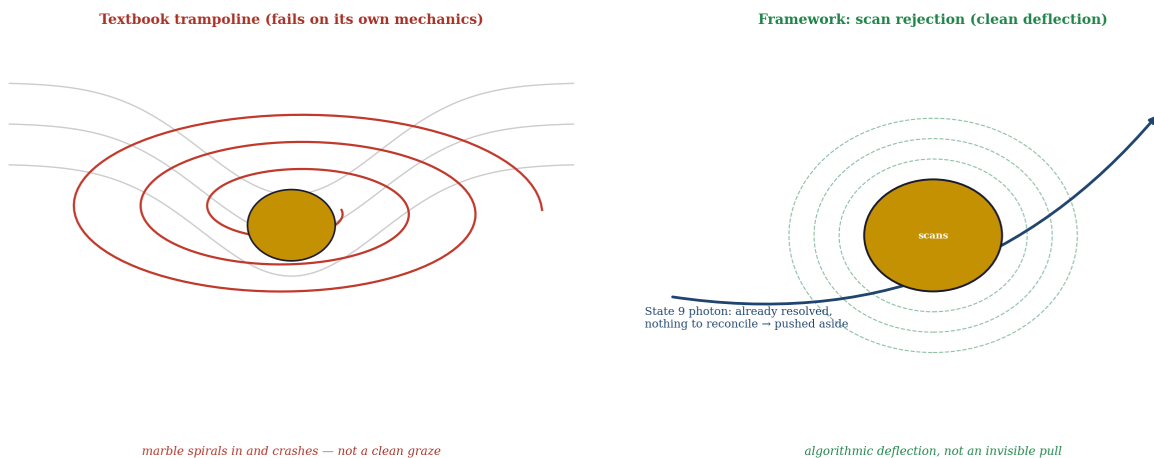


Figure 5. The textbook trampoline marble spirals in and crashes rather than grazing cleanly; the framework reads grazing as an already-resolved State 9 photon the cluster's scans cannot reconcile, deflected away as an algorithmic rejection rather than an invisible pull.

The framework removes the entire catch-22 by discarding both the invisible pull and the warped sheet, and reading the event through the scans of Papers 3 and 4.

When a photon strikes a dense cluster of merged configurations directly, the incoming geometry drives into the local matrix and forces a change of state — creation geometry, a real physical transformation. When the photon instead grazes the outer boundary without a direct hit, a different rule governs. The massive cluster is continuously executing its full manifold and temporal scans across its outer geometry, evaluating nearby configurations for ones it can reconcile with to maintain its balance. As the grazing State 9 photon passes through that boundary, the cluster's outer geometry evaluates it and finds it already a fully resolved, self-contained photon — nothing the cluster's scans can integrate or resolve against. There is no structural basis for interaction, so the outer geometry rejects it.

The rejection acts as a geometric repellent: the passing photon is deflected away from the boundary because it cannot be resolved by the local scans, not because it is attracted toward the mass. This operational deflection is what mainstream astronomy reads as gravitational curving or bent spacetime. On this account the photon is never pulled, bent, or attracted by an independent force; it is pushed aside by an active scanning rejection built into the hardware of the medium. The asymmetry of Section 5.2 dissolves, because there is no pull to switch direction: emission and grazing are simply different scan events, not two outcomes of one contested tug-of-war.

This paper uses the grazing case to expose the catch-22 and to demonstrate the scan-rejection alternative; the full observational reread of gravitational lensing, with the lensing record treated in detail, is the work of Paper 23, and is not pre-empted here.

6. The Blank Canvas: How an Adjustable Metric Manufactures Invisible Entities

The cases above share one structural root, and naming it is the purpose of this section. A theory with a single absolute baseline can run a genuine deficit: if the books do not balance against a fixed ruler, the discrepancy stands as a real tracking error demanding a structural answer. A theory free to redraw its metric at each scale cannot run such a deficit, because any imbalance can be absorbed by adjusting the local ruler or by introducing a new term sized to the gap.

The consequence follows directly. When energy or momentum fails to balance under a local, adjustable standard, the procedure that questions the standard is not available — the standard is what is being protected — so the imbalance is closed instead by positing a new entity that carries exactly the missing amount. Entities introduced this way tend to share a defining feature: they interact with the medium just weakly enough, or are confined just completely enough, that they cannot be directly registered, only inferred from the gap they were introduced to fill. A quark that can never be isolated; a neutral residue that leaves only missing energy; a mediator that appears only as a virtual term in a sum. Each closes a specific deficit, and each is shielded from direct refutation by the same property that made it useful — its inability to be directly observed.

The pattern is self-protecting, and the protection is structural rather than intentional. The measurement standard is flexible, so a secondary signal recorded by an apparatus can be read as confirmation of the template the apparatus was built around, and the core model is insulated from the kind of measurement that could break it. This is the same null-result and inference-chain concern Paper 2.5 raised for inferred particles, now traced to its origin in metrology: the entities multiply because the ruler bends. Fix the ruler — admit one absolute baseline — and the deficits that the invisible entities were introduced to absorb either

resolve into the response of the medium or stand exposed as genuine tracking errors. The framework's claim is not that every inferred entity is thereby disproved; it is that an unfalsifiable construct introduced to balance an adjustable ledger is not evidence of anything except the adjustability of the ledger.

7. The Question the Series Will Settle: Observation or Prediction

Every case in this paper reduces to a single question, and it is worth stating openly because it will recur, deliberately, through each critique paper of this series before it is taken up in full at the end. The question is this: when a direct observation and a theoretical prediction disagree, which one is permitted to be wrong?

An observable standard answers without hesitation. Water freezes where it freezes; if the prediction disagrees, the prediction is wrong. That is what it means for a standard to be public and auditable — it can overrule the template. The metrology this paper has examined answers the other way. When the prediction of a constant mass fails across scales, the mass is split in two rather than the prediction questioned (Section 3). When a simulation will not converge, a grid is imposed and tuned until it reproduces the expected answer (Section 4). When light near a star will not behave as a pulled massless object should, the rules are switched by direction of travel rather than the premise abandoned (Section 5). When a ledger will not balance, an unobservable entity is introduced to carry the remainder (Section 6). In each case the prediction is protected and the observation, or the absence of one, is made to yield.

This series does not ask the reader to accept that inversion or its opposite on assertion. It asks the reader to notice that the choice is being made at all — that a discipline which began by anchoring temperature to the visible behavior of water now lets a template overrule, absorb, or manufacture the observable as needed. One honesty is owed here: the framework's answer is a *methodological commitment*, not a derived result. Serious philosophy of science contains positions on which prediction legitimately overrules observation in periods of theory change, and this paper does not refute them; it takes a side, openly, exactly as the framework takes three axes and three values as its starting hardware. The commitment is observational primacy: observation outranks prediction, a standard earns authority by being checkable, and a physics built on an observable baseline does not need to manufacture what it cannot see. The argument of this paper is complete on that commitment's own terms — the cases stand as documented, and the reader who grants the commitment has the full indictment in hand here, not in a later installment. What the capstone companion to the Compendium adds is depth, not the missing argument: the case-by-case accounting across all the critique papers, and the methodological question engaged as philosophy in its own right. This paper plants the question and answers it for its own cases; the capstone widens the ledger.

8. Conclusion: Re-Anchoring the Metrics of Nature

The audit disputes none of the measured laboratory quantities. Charges register as integers; nucleon masses are what they are; light deflects near mass as observed; the Sun sheds the mass it sheds. What the audit disputes is the metrology laid over those measurements: a system that, lacking an absolute zero, assigns a different ruler to each flavor, generation, and scale; that builds the same fragmentation into its computational core, where a grid must be imposed, balanced by decree, tuned by hand, and extrapolated away; that switches the rules governing light by its direction of travel; and that, with a metric it can redraw

at will, absorbs every tracking error into an entity defined by its inability to be seen.

The framework's alternative is the one Paper 4 set: the photon as the sovereign zero, carrying no surface departure and running only the inner identity scan; the electron-positron pair as the first resolved mark on a single continuous three-axis fabric, reached by three conserving tilts and fixed by the Per-Triangle Rule with no tunable scale. On that one baseline, symmetries hold without a software decree, mass is accounted without running constants, and the trajectories of light are resolved without an invisible pull. Space is returned to a finite, auditable geometric medium whose ledger cannot be balanced by adjusting the coordinate system — because there is no coordinate system to adjust, only the medium and its departures from baseline. The standard the series returns to is the one the Celsius lesson named: a baseline you can check, against which a prediction is not permitted to be the thing that wins.

References

This paper is the critique companion to Paper 4 and part of the *Departure From the Photon Baseline* series. It cites the companion papers below by their published records, and the primary historical sources for the methodological history of Section 1.

1. Kadunic, D. (2026). *The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. <https://doi.org/10.5281/zenodo.20434286>
2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. <https://doi.org/10.5281/zenodo.20510019>
3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. <https://doi.org/10.5281/zenodo.20518514>
4. Kadunic, D. (2026). *The Nine-State Classification with Inside-Outside Structure* (Paper 3). Zenodo. <https://doi.org/10.5281/zenodo.20561941>
5. Kadunic, D. (2026). *The Constant That Does Not Stay Constant: A Structural Audit of the Fine-Structure Measurement* (Paper 3.5). Zenodo. <https://doi.org/10.5281/zenodo.20575611>
6. Kadunic, D. (2026). *The Photon as Baseline, Electron-Positron as First Tilt* (Paper 4). Zenodo. [companion to this paper]
7. Dirac, P. A. M. (1978). *Directions in Physics*. Wiley. (Source of the "not sensible mathematics" objection to renormalization, from his 1975 lecture.)
8. Feynman, R. P. (1985). *QED: The Strange Theory of Light and Matter*. Princeton University Press. (Source of the "dippy process" and "hocus-pocus" remarks on renormalization, cited in Paper 3.5.)
9. Unruh, W. G. (1976). "Notes on black-hole evaporation." *Physical Review D*, 14(4), 870. (The predicted acceleration-dependent particle bath; never directly observed.)

Paper 4 establishes the sovereign zero, the first resolved mark, and the Per-Triangle Rule's scale invariance that this paper contrasts against standard metrology. Paper 2.5 establishes the inference-chain and null-result concerns that Section 6 traces to their origin in metrology. The full positive account of composite configurations (Section 3) is the work of Paper 16; the full reread of gravitational lensing (Section 5) is the work of Paper 23; the photon-photon scattering case noted in development is reserved for the future high-energy-collisions paper, where the recent direct-observation data can be engaged in detail; the observation-versus-prediction question (Section 7) is answered here for this paper's cases and deepened across the series in the capstone companion to the Compendium. None is pre-empted here.

A. The Fragmented-Metric Translation Ledger

The table maps the standard model's scale-dependent measurement devices to their framework readings, in the manner of the g-2 ledger of Paper 3.5, Appendix A.

Standard device	Standard definition	Framework reading
Current quark mass	Bare mass parameter in the core equations	One scale-reading of a continuous junction under low probe load
Constituent quark mass	Effective mass including binding energy, ~100x larger	The same junction read under a different load, relabeled as a different mass
Running coupling constant	Interaction strength that varies with energy scale	The fabric's load-dependent response, recorded as if the constant itself changed
Fractional charge (1/3, 2/3)	Intrinsic charge of a confined constituent	A bookkeeping fraction of one continuous junction; registered by instruments only as integers
Lattice gauge invariance	Symmetry constraint keeping grid calculations stable	Stability imposed on a calculation by decree, in place of the medium's own Per-Triangle balance
Grid spacing and the continuum limit	Numerical parameter, removed by extrapolation to zero	A tunable scale the framework does not contain; nothing to tune or extrapolate
Tuning to the physical point	Calibrating inputs to reproduce a known measured value	Fitting the scale by hand to a baseline the framework fixes absolutely
Renormalized self-energy	Finite mass recovered by subtracting a divergence	An infinity that never arises, because the medium has no zero-size point and a finite response

B. The Quark Mass Ledger as a Worked Case of Targeted Ruling

This appendix sets the quark mass figures side by side as a single audit, to show the fragmentation as a matter of the field's own numbers rather than of characterization.

Consider the proton, a composite the standard model builds from two up quarks and one down quark. The bare current masses assigned to those three quarks sum to a small fraction — under two percent — of the proton's measured mass. The remaining mass is attributed to the energy of the binding field. To carry the three quarks as the carriers of the proton's mass, the treatment then switches to constituent masses inflated by a factor near one hundred, so that the three constituents now sum to approximately the measured total.

The structural feature to note is that the same three objects are assigned two mass values differing by two orders of magnitude, and which value is used is decided by which quantity must be reproduced. When the core interaction is computed, the small current masses are used. When the composite weight must be matched, the large constituent masses are used. No single mass is carried consistently through both calculations.

Across generations the pattern repeats without a unifying law. Each of the six flavors is assigned its own mass parameter, and those parameters span many orders of magnitude — from the light up and down quarks to a top quark weighed at the scale of a whole gold atom. There is no derivation from which the six values follow; each is fixed to its own measurements. The audit's finding is the same as the body's: this is a catalog of local standards, not a single ruler reading a single medium. The framework's positive account — one junction, one medium, the apparent flavors as load-dependent fractures of it — is developed in Paper 16 and is not argued here; this appendix only lays out the standard ledger so the reader can see the shifting ruler in the numbers.

C. The Two-Premise Contradiction in the Standard Deflection Account

This appendix states the relativistic catch-22 of Section 5.1 as a side-by-side inconsistency, so the contradiction can be read directly rather than narrated.

Premise	What it asserts	What it requires of the photon near mass
Velocity escape	The photon escapes capture because it moves at c , fast enough to clip the well and leave	That its speed <i>reduces</i> its effective susceptibility to capture
Mass-energy of motion	As velocity approaches c , energy density and momentum grow without bound	That its speed <i>increases</i> the energy and momentum that source gravitational interaction
Stress-energy sourcing	Energy density and momentum gravitate as mass does	That the photon's growing energy/momentum <i>strengthens</i> its gravitational interaction

The three rows cannot all hold for the same photon at once: the first requires speed to weaken capture, while the second and third require that same speed to strengthen the interaction that produces capture. The standard account keeps all three by treating the deflection as the small residual of a near-escape, but the residual is computed within the very framework whose premises conflict. The framework's resolution, given in Section 5.4, removes the contradiction at its root: the photon is neither pulled nor curved; it is an already-resolved State 9 that a massive cluster's scans cannot reconcile and therefore deflect away. The detailed observational reread belongs to Paper 23.

D. The Direction-Dependent Bookkeeping of Stellar Radiation

This appendix lays the stellar mass-loss paradox of Section 5.2 out as a two-direction ledger, so the asymmetry can be inspected directly.

Photon's direction of travel	Standard model's treatment	What that treatment requires
Leaving the star (emission, inside to out)	Radiated freely; the star sheds ~4 million metric tons per second as light	That the gravity well does <i>not</i> contain the photon, even from the point of deepest field
Grazing the star (passing, outside)	Bent inward by the gravity well; the basis of lensing	That the gravity well <i>does</i> act on the photon strongly enough to curve its path

The same gravitational field, acting on the same kind of massless entity, is required to release it in one direction of travel and capture-and-bend it in the other. Gravitational redshift is offered as the cost of leaving, but it adjusts the photon's energy rather than removing the asymmetry: a mass-driven interaction is applied to an entity defined as massless, and the outcome is allowed to differ by direction. The framework's resolution removes the asymmetry entirely (Section 5.4): there is no pull to contain or release the photon. Emission is a creation-geometry event at the source; grazing is a scan-rejection event at a distant boundary. Two different scan events, not two contradictory outcomes of one gravitational tug-of-war.